

Unit: 4

- Sum rule ($m+n$)
- Multiplication rule ($m \times n$)
- Permutations
- Combination
- Factorial

* Basics of Counting

Counting in PM helps us answering the questions while ^{Q1)} how many ways can an event occur

Q2) How many arrangements or selections are possible

Q3) Mainly the basics includes 4 principles

(1) Addition (sum) / Disjunctive rule

If an event occurs in m ways and another event occurs in n ways and if these 2 events cannot occur simultaneously

Ex: If there are 5 boys and 4 girls in a class, then there are $5+4=9$ ways of selecting one student (either a boy or a girl) as CR.

(2) Product Rule

If an event occurs in m ways and 2nd event can occur in n ways and if the no. of ways the 2nd event occurs does not depend upon

how the first event occurs and the 2 events can occur, simultaneously in $n \times n$ ways.

→ Exr If 2 dices are rolled, then the first dice can fall in 6 ways as event 1 and 2nd dice can fall in 6 ways as event 2. Hence, there are $E_1 \times E_2 = 36$ ways when 2 dices are rolled

→ Factorial $n! = n(n-1)(n-2) \dots 3 \times 2 \times 1$

$n!$ (or) LD

Exr A I R can be represented in $3!$ ways
(AIR, IAR, RIA, RAI, IRA, ARI)

Exr RAMANA

It can be represented by $\frac{6!}{3!} = \frac{6 \times 4 \times 5 \times 3!}{3!}$

Exr COMMITTEE = $\frac{9!}{2!2!2!} = \frac{9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{2 \times 2 \times 2} = \frac{9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{2 \times 2 \times 2} = 8 \times 4 \times 2 \times 5 \times 9 = 72 \times 5 \times 42 = 360 \times 42 = 72 \times 42 = 4920$

Permutation

- This is an advanced counting method. It is an arrangement of an object of out of n different positions. r element

Can be arranged in nPr ways. Without repetition = $\frac{n!}{(n-r)!}$

With repetition = n^r

→ 4 $n=3, r=2$

Without rept	<u>A</u> <u>B</u> <u> </u>		<u>B</u> <u>A</u> <u> </u>	= $\frac{3!}{(3-2)!}$ = 6 //
	<u> </u> <u>B</u> <u>A</u>		<u> </u> <u>A</u> <u>B</u>	
	<u>A</u> <u> </u> <u>B</u>		<u>B</u> <u> </u> <u>A</u>	

With rept $n=3, r=2$

$$= 3^2 = 9$$

② ② ② (A B)

$n=2$ = No. of people to occupy
 r = No. of arrangements = 3

<u>A</u> <u>A</u> <u>A</u>		<u>B</u> <u>A</u> <u>B</u>
<u>B</u> <u>B</u> <u>B</u>		<u>A</u> <u>B</u> <u>B</u>
<u>A</u> <u>A</u> <u>B</u>		<u>B</u> <u>B</u> <u>A</u>
<u>A</u> <u>B</u> <u>A</u>		<u>B</u> <u>A</u> <u>A</u>

→ Combinations

Combination is a selection out of n different elements, r elements can be selected in nCr ways.

Without repetition = $\frac{n!}{r!(n-r)!}$

With repetition = $(n+r-1)C_r$

3 elements

Exr Committee instructs a committee of 6 members must be formed from 7 men and 5 women atleast 2 women must be included : how many different committees can be formed.

$$\begin{array}{l} \rightarrow \text{Com} = 6 \\ M = 7 \\ W = 5 \end{array} \quad \left. \begin{array}{l} 2W + 4M \\ 3W + 3M \\ 4W + 2M \\ 5W + 1M \end{array} \right\} \rightarrow$$

$$= {}^5C_2 * {}^7C_4 + {}^5C_3 * {}^7C_3 + {}^5C_4 * {}^7C_2 + {}^5C_5 * {}^7C_1$$

$$= 350 + 350 + \dots = 812 \text{ ways}$$

(Q) A box contain 2 white, 3 black and 4 ^{red} balls. In how many ways can 3 balls be drawn if atleast one black is included.

$$1b + {}^8C_2$$

$$1b \dots$$

$$= {}^8C_2 \times {}^3C_1 = 84$$

$$\begin{array}{l} {}^3C_1 * {}^6C_2 \\ {}^3C_2 * {}^6C_1 \\ {}^3C_3 * {}^6C_0 \end{array}$$

$${}^3C_1 = 1 \text{ black ball}$$

$$= \text{Other 2 not balls}$$

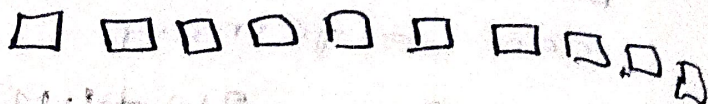
$$\begin{array}{rcl} 3 * 15 & = & 45 \\ 3 * 6 & = & 18 \\ 1 * 1 & = & 1 \\ \hline & & 64 \end{array}$$

$$\text{(or)} \quad {}^9C_3 - {}^6C_3$$

In how many ways are there to distribute 10 different books among 15 people if no person is received more than 1 book.

$$15 = n$$

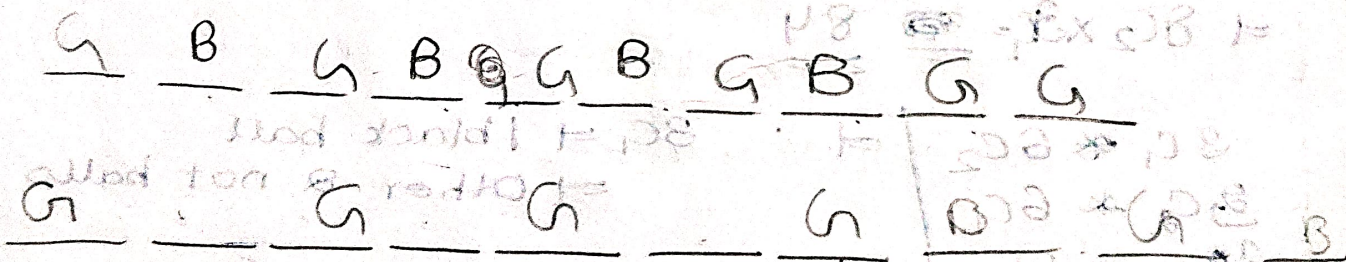
$$10 = r$$



15 people

$$\begin{aligned} {}^{15}P_{10} &= \frac{15 \times 14 \times 13 \times 12 \times 11 \times 10 \times 9 \times 8 \times 7 \times 6}{1} \\ &= \frac{15!}{5!} \\ &= 15 \times 14 \times 13 \times 12 \times 11 \times 10 \times 9 \times 8 \times 7 \times 6 \end{aligned}$$

Find the no. of different ways in which 4 boys and 6 girls are such that no 2 boys are together.



$${}^6P_6 \times {}^4P_4$$

$$= 1 \times 24 = 24$$

$$= \boxed{604800}$$

→ When do we use combinations with repetitions?

The number of r combinations of n distinct objects with unlimited repetitions

The no. of non-negative integral solutions

$$x_1 + x_2 + \dots + x_k = r$$

The number of ways of distributing r similar balls into N numbered boxes.

The no. of binary numbers with $(n-1)$ ones and r zeros then $c(n-1+r, r)$

$$= \boxed{n-1+r C_r}$$

(Q1) In how many ways can we distribute 7 apples, 6 oranges among 4 children so that each child gets 1 apple.

Sol

$$\frac{1}{1C} \quad \frac{1}{2C} \quad \frac{1}{3C} \quad \frac{1}{4C}$$

If 4 apples are already distributed to 4 children then 3 apples remain

$$\left. \begin{array}{l} n = 4 \\ r = 3 \end{array} \right\} \text{ Apples} \quad \left. \begin{array}{l} n = 4 \\ r = 6 \end{array} \right\} \text{ Oranges}$$

↳ No. of children.

$$= 9C_4$$

$$\boxed{6C_3 \times 9C_6}$$

$$= \cancel{6 \times 5 \times 4} \times \cancel{9 \times 8 \times 7 \times 6}$$

$$= \cancel{20 \times 6 \times 63}$$

$$= \cancel{120 \times 63}$$

$$= \underline{\underline{7560}}$$

$$\begin{array}{r} 12 \\ \times 63 \\ \hline 36 \\ 72 \times \\ \hline 756 \end{array}$$

$$\frac{6 \times 5 \times 4 \times 3!}{3!} \times \frac{9 \times 8 \times 7 \times 6!}{6!} = 1680$$

If 1 apple is given to each student then

apples $\Rightarrow 3 \longrightarrow 4$ students

$$n+r-1C_r$$

$$4+1+3C_3 = 6C_3$$

Oranges $\Rightarrow 6 \longrightarrow 4$ students

$$n+r-1C_r = \underline{\underline{9C_3}}$$

$$\therefore \text{Total no. of ways} = \text{No. of ways of ap} \times \text{No. of ways of or}$$
$$= 680$$

(Q) Find the no. of positive integer solutions of the equations $x_1 + x_2 + x_3 = 17$ where $x_1 \geq 1$
 $x_2 \geq 1$; $x_3 \geq 1$

$x_1 = \text{ways}$

$\therefore$ By conditions we can know $x_3 \leq 16$

lower Boundness

$$x_1 \geq 1, x_2 \geq 1, x_3 \geq 1$$

$$x_1 + x_2 + x_3 = 3 \text{ (min sum)}$$

$$\text{(Max sum)} = 17$$

Sol Let us take y_1, y_2, y_3

$$y_1 = x_1 - 1 \Rightarrow x_1 = y_1 + 1$$

$$y_2 = x_2 - 1 \Rightarrow x_2 = y_2 + 1$$

$$y_3 = x_3 - 1 \Rightarrow x_3 = y_3 + 1$$

$$\therefore (y_1 + 1) + (y_2 + 1) + (y_3 + 1) = 17$$

$$y_1 + y_2 + y_3 = 14$$

We take $n = 3$; $r = 14$

$$\Rightarrow \boxed{n+r-1 \text{ } ^nC_r}$$

$$\Rightarrow 17-1 \text{ } ^{16}C_4 = \underline{16C_4}$$

$$\frac{16!}{14! 2!} = 15 \times 8$$

$$= \boxed{120}$$

(Q) Find the number of +ve integer solutions

$$x_1 + x_2 + x_3 + x_4 + x_5 \geq 30$$

$$x_1 \geq 2$$

$$x_4 \geq 2$$

$$x_2 \geq 3$$

$$x_5 \geq 0$$

$$x_3 \geq 4$$

sol Let us take y_1, y_2, y_3, y_4, y_5

$$y_1 = x_1 - 2 \Rightarrow x_1 = y_1 + 2$$

$$y_2 = x_2 - 3 \Rightarrow x_2 = y_2 + 3$$

$$y_3 = x_3 - 4 \Rightarrow x_3 = y_3 + 4$$

$$y_4 = x_4 - 2 \Rightarrow x_4 = y_4 + 2$$

$$y_5 = x_5 - 0 \Rightarrow x_5 = y_5$$

$$\Rightarrow (y_1 + 2) + (y_2 + 3) + (y_3 + 4) + (y_4 + 2) + y_5 = 30$$

$$\Rightarrow y_1 + y_2 + y_3 + y_4 + y_5 = 30 - 11$$

$$= 19$$

$$\Rightarrow n = 5 \quad r = 19$$

$$\Rightarrow {}^{23}C_{19} = \frac{23 \times 22 \times 21 \times 20}{4 \times 3 \times 2 \times 1}$$

$$= 35 \times 11 \times 23$$

$$= \boxed{8855}$$

(Q) Find the no. of non-negative integers x_1, x_2, x_3 such that $x_2 + x_3 = 6$

$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0$$

$$\therefore x_1 + x_2 + x_3 = 6$$

$$n = 3$$

$$r = 6$$

$$\therefore \text{No. of ways} = 6 + 3 - 1 {}^6C_6$$

$$= {}^8C_6 = \boxed{28}$$

(Q) The no. of binary nos with 10 ones and 5 zeros

$$r = 5$$

$$\text{Then } \frac{C(n-1+r, r)}$$

$$10 \text{ ones} = n-1$$

$$\text{No. of ones} = n-1$$

$$\text{No. of zeros} = r$$

$$\Rightarrow n = 11$$

$$\Rightarrow (n+r-1) {}^nC_r = \boxed{{}^{15}C_5}$$

$$= \frac{15 \times 14 \times 13 \times 12 \times 11}{5 \times 4 \times 3 \times 2 \times 1}$$

$$= \frac{3 \times 7 \times 11 \times 10}{1 \times 2 \times 1} = 21 \times 11 \times 5$$

$$= \boxed{3003}$$

Find the no. of distinct order triple of x_1, x_2, x_3 of non negative integers satisfying the inequality

$$x_1 + x_2 + x_3 < 6$$

Sol $\therefore x_1 + x_2 + x_3 \leq 5$

$$x_1 + x_2 + x_3 = 1 \Rightarrow n = 3, r = 1 \Rightarrow 3 + 1 - 1 C_1 = 3C_1$$

$$x_1 + x_2 + x_3 = 2 \Rightarrow n = 3, r = 2 \Rightarrow 3 + 2 - 1 C_2 = 4C_2$$

$$x_1 + x_2 + x_3 = 3 \Rightarrow n = 3, r = 3 \Rightarrow 3 + 3 - 1 C_3 = 5C_3$$

$$x_1 + x_2 + x_3 = 4 \Rightarrow n = 3, r = 4 \Rightarrow 3 + 4 - 1 C_4 = 6C_4$$

$$x_1 + x_2 + x_3 = 5 \Rightarrow n = 3, r = 5 \Rightarrow 3 + 5 - 1 C_5 = 7C_5$$

$$x_1 + x_2 + x_3 = 0 \Rightarrow n = 3, r = 0 \Rightarrow 3 + 0 - 1 C_0 = 1C_0$$

$$\Rightarrow 3C_1 + 4C_2 + 5C_3 + 6C_4 + 7C_5 + 1C_0$$

$$\Rightarrow \boxed{56}$$

Find the no. of binary no.s with 2 ones and 11 zeros

$$n - 1 = 21 \quad r = 11$$

$$n = 22$$

$$\Rightarrow \text{No. of nos} = 22 + 11 - 1 C_{11}$$

$$= 32C_{11}$$

$$= \underline{\underline{129024480}}$$

→ (Q) Find the no. of distinct no.s x_1, x_2, x_3, x_4 non negative integer $x_1 + x_2 + x_3 + x_4 < 9$

$$x_1 + x_2 + x_3 + x_4 \leq 8$$

$$x_1 + x_2 + x_3 + x_4 = 0 \quad \left| \begin{array}{l} n = 4 \\ r = 0 \end{array} \right| \quad 3C_0$$

$$x_1 + x_2 + x_3 + x_4 = 1 \quad \left| \begin{array}{l} n = 4 \\ r = 1 \end{array} \right| \quad 4C_1$$

$$x_1 + x_2 + x_3 + x_4 = 2 \quad \left| \begin{array}{l} n = 4 \\ r = 2 \end{array} \right| \quad 5C_2$$

$$x_1 + x_2 + x_3 + x_4 = 3 \quad \left| \begin{array}{l} n = 4 \\ r = 3 \end{array} \right| \quad 6C_3$$

$$x_1 + x_2 + x_3 + x_4 = 4 \quad \left| \begin{array}{l} n = 4 \\ r = 4 \end{array} \right| \quad 7C_4$$

$$\begin{array}{l|l|l}
 x_1 + x_2 + x_3 + x_4 = 5 & n=4 & r=5 \\
 x_1 + x_2 + x_3 + x_4 = 6 & n=4 & r=6 \\
 x_1 + x_2 + x_3 + x_4 = 7 & n=4 & r=7 \\
 x_1 + x_2 + x_3 + x_4 = 8 & n=4 & r=8
 \end{array}
 \begin{array}{l}
 8C_5 \\
 9C_6 \\
 10C_7 \\
 11C_8
 \end{array}$$

Total no. of ways =

$$3C_0 + 4C_1 + 5C_2 + 6C_3 + 7C_4 + 8C_5 + 9C_6 + 10C_7 + 11C_8$$

$$= \boxed{495}$$

→ Permutations with Constraints & Repetition

* The no. of permutations of n letter word which R_i are of one type $P(n_1, n_2, n_3, \dots, n_k)$ then

$$\frac{n!}{r_1! r_2! r_3! \dots r_k!}$$

Ex Find the no. of arrangements of the word "Engineering"

n = No. of letters in a word

$$n = 11$$

$$E = 3$$

$$N = 3$$

$$G = 2$$

$$I = 2$$

$$\Rightarrow \text{Total no. of arrangements} = \frac{n!}{r_1! r_2! r_3! \dots}$$

$$= \frac{11!}{3! 3! 2! 2! 1!}$$

$$= 11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4!$$

Ex MISSISSIPPI

$$M = 1$$

$$I = 4$$

$$S = 3$$

$$P = 2$$

$$\Rightarrow \frac{11!}{n!}$$

$$11!$$

$$4! 4! 2!$$

$$= 11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4!$$

$$= 4! \times 2! = 34600$$

TALLA HASSEEE

$$\begin{aligned} T &= 1 & S &= 2 \\ A &= 3 & E &= 2 \\ L &= 2 \\ H &= 1 \end{aligned}$$

$$\begin{aligned} \text{No. of ways} &= \frac{11!}{2!2!2!3!} \\ &= \frac{11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{2 \times 2 \times 2 \times 3 \times 3 \times 3} \\ &= 15 \times 8 \times 6 \times 9 \times 2 \\ &= 8731600 \end{aligned}$$

Binomial coefficient is defined as $nC_r = \frac{n!}{(n-r)!r!}$

It represents the no. of ways to choose r objects from n objects so, it also appears in Combinatorics

Binomial Theorem

If n is a positive integer then

$$(x+y)^n = \sum_{r=0}^n nC_r x^{n-r} y^r$$

(Observation)

The expansion of $(x+y)^n$ contains $(n+1)$ terms. The $(r+1)$ th term of the expansion of $(x+y)^n$ is $nC_r x^{n-r} y^r$

The expansions of $(x+y)^n$ and $(y+x)^n$ are equal

Ex Find the middle term of $\left(2x + \frac{1}{39x}\right)^{10}$

$(x+y)^n$ contains $(n+1)$ terms

$\therefore$ 11 terms

Middle = 6th term

$$n=10$$

$$\therefore \text{Middle term} = {}^{10}C_5 (2x)^5 \left(\frac{-1}{3x}\right)^5$$

$$= {}^{10}C_5 (32x^5) \left(\frac{-1}{3^5x^5}\right)$$

$$= -1 \left(\frac{10 \times \cancel{9} \times 8 \times 7 \times \cancel{6} \times \cancel{5} \times 1 \times 3 \times 2}{\cancel{9}! \times \cancel{9} \times \cancel{9} \times 3} \right)$$

$$= - \left(\frac{80 \times 7 \times 32}{9} \right)$$

$$= - \left(\frac{1920}{9} \right) = -33.18$$

$$\rightarrow *) \left(x - \frac{3}{y}\right)^9$$

$$\text{Middle term } r=5 \text{ (or) } 6$$

$$\boxed{\begin{matrix} n=9 \\ r=5 \end{matrix}}$$

$$= {}^nC_r (x)^{n-r} \left(\frac{-3}{y}\right)^r$$

$$= {}^9C_5 (x)^4 \left(\frac{-3}{y}\right)^5$$

$$= - {}^9C_5 x^4 \frac{3^5}{y^5} = \frac{9! \cdot 3^5}{5! \cdot 4!} \left(\frac{x^4}{y^5}\right)$$

$$= \frac{9 \times \cancel{8} \times 7 \times \cancel{6} \times \cancel{5} \times 3^5}{\cancel{8}! \cdot \cancel{4}!} \left(\frac{x^4}{y^5}\right)$$

$$= 18 \times 7 \times 3^5 \left(\frac{x^4}{y^5}\right)$$

$$\boxed{-30618 \left(\frac{x^4}{y^5}\right)}$$

$$n=9$$

$$r=6$$

$$(r+1)\text{th term} = 6$$

$$(r+1)\text{th term} = 5$$

$$r=5$$

$$r=4$$

$$= {}^9C_4 (x)^5 \left(-\frac{3}{y}\right)^4$$

$$= {}^9C_4 x^5 \times \frac{3^4}{y^4} =$$

$$= \frac{9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{4! \times 5!} \times 3^4 \left(\frac{x^5}{y^4}\right)$$

$$= 11 \times 9 \times \frac{x^5}{y^4} = \boxed{126 \frac{x^5}{y^4}}$$

$$*) \left(\frac{x}{2} + \frac{3}{x}\right)^{11}$$

⇒ Middle term + 6th term

$$(r+1)\text{th term} = 6$$

$$r=5$$

$$= {}^{11}C_5 \left(\frac{x}{2}\right)^6 \left(\frac{3}{x}\right)^5$$

$$= \frac{11!}{5! 6!} \left(\frac{x^6}{2^6}\right) \left(\frac{3^5}{x^5}\right)$$

$$= \frac{11 \times 10 \times 9 \times 8 \times 7 \times 6!}{6! \times 5!} \times \frac{3^5}{2^6}$$

$$= \frac{12 \times 11 \times 10 \times 9 \times 8 \times 7}{12 \times 11 \times 10 \times 9 \times 8 \times 7} \times \frac{3^5}{2^6} =$$

⇒ Middle term + 7th term $(r+1) = 7$

$$r=6$$

$$= {}^{11}C_6 \left(\frac{x}{2}\right)^5 \left(\frac{3}{x}\right)^6$$

$$= \frac{11 \times 10 \times 9 \times 8 \times 7 \times 6!}{6! \times 5!} \times \frac{3^6}{2^5} \times \frac{1}{x} = 22 \cdot 781 \times \frac{1}{x}$$

$$= 10,524 \cdot 82 \frac{2}{x}$$

Find the coefficient of $x^9 y^3$ in the expansion of $(2x - 3y)^{12}$

Solr $x^9 y^3$

By Binomial theorem

$$(x+y)^n = \sum_{r=0}^n nC_r x^{n-r} y^r$$

$$(2x - 3y)^{12} = \sum_{r=0}^{12} {}^{12}C_r x^{12-r} y^r$$

$$n=12,$$

$$x^9 = x^{12-r} \quad \boxed{r=3}$$

Term

$$= {}^{12}C_3 (2x)^9 (-3y)^3$$

$$= \frac{12 \times 11 \times 10 \times 9!}{3! \times 9!} 2^9 (-27) x^9 y^3$$

$$= 220 \times 2^9 \times -27$$

$$= \underline{\underline{-3041280 (x^9 y^3)}}$$

Find the term independent of x in the expansion of $(x^2 + \frac{1}{x})^{12}$

$$(x^2 + \frac{1}{x})^{12}$$

$$(x^2)^{n-r} \left(\frac{1}{x}\right)^r$$

$$= \frac{x^{2n-r}}{x^r} = 1$$

$$2n-r=r$$

$$2n=2r$$

$$\boxed{n=r}$$

$$nC_r x^{n-r} y^r \quad \text{(15th)th term.}$$

$$= {}^{12}C_r x^{12-r} y^r$$

$$= {}^{12}C_r x^{24-2r} \left(\frac{1}{x}\right)^r$$

The power of $x=0$

$$= {}^{12}C_r x^{24-2r-r} = 0$$

$$24-2r-r=0$$

$$\boxed{r=8}$$

$$= {}^{12}C_8 (x^2)^4 \left(\frac{1}{x}\right)^8$$

$$= \boxed{495}$$

Q) Find the 2 successive terms in the expansion of $(1+x)^7$ where coefficients are in the ratio of 3:5

Sol $(1+x)^7$; 3:5

$r, r+1$

$$\frac{T_r}{T_{r+1}} = \frac{r+1}{n-r} = R$$

$$(x+y)^n = {}^nC_r x^{n-r} y^r$$

$$n=7$$

$$r=2$$

$$\frac{T_2}{T_3} = \frac{r+1}{n-r} = \frac{3}{5}$$

$$\Rightarrow 5r+5 = 21-3r$$

$$\Rightarrow 8r = 16$$

$$\Rightarrow \boxed{r=2}$$

$$\therefore \frac{{}^nC_r}{{}^nC_{r+1}} = \frac{{}^nC_2}{{}^nC_3}$$

$$T_2 = T_2 ; T_{r+1} = T_3$$

Note

① If we take $y=1$ in the expansion of $(y+x)^n$ we have $(1+x)^n = C(n,0) + C(n,1)x + C(n,2)x^2 + \dots + C(n,r)x^r + \dots + C(n,n)x^n$

* The Expansion of $(y-x)^n$ is $[C(n,0)y^n - C(n,1)y^{n-1}x + C(n,2)y^{n-2}x^2 - \dots + (-1)^r C(n,r)y^{n-r}x^r + \dots - (-1)^n x^n]$

* If we take $y=1$ in the expansion of $(y-x)^n$ we get $(1-x)^n = C(n,0) - x C(n,1) + x^2 C(n,2) - x^3 C(n,3) + \dots + (-1)^n x^n$

Properties of Binomial Coefficients

① Prove that $C(n,0) + C(n,1) + C(n,2) + \dots + C(n,n)$
 $= 2^n$

Soln $(y+x)^n = \sum_{r=0}^n nC_r y^{n-r} x^r$

$y=1$

$$(1+x)^n = \sum_{r=0}^n nC_r \cdot (1)^{n-r} x^r$$

$$(1+x)^n = C(n,0) + C(n,1)x + C(n,2)x^2 + \dots + C(n,n)x^n$$

Put $x=1$:

$$(1+1)^n = C(n,0) + C(n,1) + C(n,2) + \dots + C(n,n)$$

$$2^n = C(n,0) + C(n,1) + C(n,2) + \dots + C(n,n)$$

② PT $C(n,0) + C(n,2) + C(n,4) + \dots$
 $= C(n,1) + C(n,3) + C(n,5) + \dots$

$$(1+x)^n = C(n,0) + C(n,1)x + C(n,2)x^2 + \dots = 2^n$$

Put $x=-1$:-

$$0^n = C(n,0) - C(n,1) + C(n,2) - C(n,3) + \dots$$

$$C(n,0) + C(n,2) + C(n,4) - C(n,1) - C(n,3) - \dots = 0^n - 0$$

We have

$$C(n,0) + C(n,2) + C(n,4) + \dots = 2^{n-1}$$

terms

∴ We have to prove the sum of even indexed terms

Let $x=1$ $(1+1)^n = C(n,0) + C(n,1) + C(n,2) + \dots$

$$\underline{\underline{x = -1}} \Rightarrow c(n,0) - c(n,1) + c(n,2) - c(n,3) \dots$$

$$\text{Add} = 2 \times 101 = 2(c(n,0) + c(n,2) + c(n,4) \dots)$$

$$\boxed{\underline{\underline{2^{n-1}}} = c(n,0) + c(n,2) + c(n,4) \dots}$$